SUNG RYE PARK

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PROFESSIONAL SUMMARY

Computational biologist specializing in **target prioritization, multi-omics integration, and interpretable machine learning** for drug discovery. Translates complex multi-modal data into actionable target hypotheses and preclinical decisions — across immune, cardio-metabolic, and neurodegenerative disease areas. Track record of delivering pharmaceutical-grade translational analyses at Battelle and Dana-Farber — including direct BMS collaboration linking clinical treatment response (responder/non-responder) to immune cell dynamics via scRNA-seq/scTCR-seq. Domain experience spans immuno-oncology and metabolic disease (hepatic transcriptomics, high fat diet models).

EDUCATION

Ph.D. in Plant Biology (Epigenetics) | University of Texas at Austin, Austin, Texas | 2010 – 2017
M.S. in Biology (Education) | Seoul National University, Seoul, Korea | 2007 – 2010
B.S. in Biology | Yonsei University, Seoul, Korea | 2005 – 2007

PROFESSIONAL EXPERIENCE

DATA SCIENTIST II, BIOINFORMATICS | BATTELLE, COLUMBUS, OH | 2025 – PRESENT

Health Analytics Division

Independently defining and driving bioinformatics strategy for pharmaceutical collaborations and government research programs, delivering multimodal integration frameworks and production-ready analytical infrastructure.

VIRTUAL PATIENT COHORT GENERATION – PROJECT LEAD (NOV 2025 – PRESENT)

Independently scoped and leading development of a CVAE framework to generate synthetic patient populations stratified by disease status (TCGA LUAD, expanding to TCGA COAD). Integrates public genomic databases with drug response profiles to enable in silico therapeutic outcome prediction — supporting preclinical go/no-go decision frameworks.

GOVERNMENT RESEARCH PROGRAMS (SEP 2025 – PRESENT)

Designing and validating a multi-omics modeling platform integrating six genomic modalities (RNA-seq, ATAC-seq, methylation, variant calling, splicing, proteomics). Benchmarking multiple integration approaches against synthetic and public datasets — with systematic parameter tuning to maximize the breadth and reliability of detectable target pools prior to production deployment.

JAPAN TOBACCO COLLABORATION: MULTI-MODAL DRUG SCREENING (JUN 2024 – PRESENT)

Built integrated analysis framework combining Cell Painting phenotypic profiling with transcriptomic data across 128 compounds to enable rapid compound prioritization via structure-based clustering. Evaluated and benchmarked multi-omics integration methods (DIABLO, MOFA2, custom designs) — partner selected and independently extended MOFA2 into a novel combinatorial framework (SOFA), with groundwork credited as co-authorship at EuroTox 2026.

ADPANEL: ALZHEIMER'S DISEASE RESEARCH – PROJECT LEAD (APR – SEP 2025)

Designed and delivered a scalable data integration system harmonizing public RNA-seq datasets through custom batch correction and canonical correlation methods. Built version-controlled pipeline linking AD transcriptional signatures with environmental exposure patterns, with full documentation for cross-team handoff.

SENIOR BIOINFORMATICS SCIENTIST | DANA-FARBER CANCER INSTITUTE, BOSTON, MA | 2022 – 2024

Informatics & Analytics, Bioinformatics Group — Pharmaceutical Partnerships

Embedded computational biologist supporting five pharmaceutical programs across oncology (NSCLC, SCLC, PTCL) — translating partner scientific questions into analytical decisions and deliverables. Contributed to experimental design as well as analysis, closing the gap between wet lab and computational teams.

REPRESENTATIVE PHARMACEUTICAL COLLABORATIONS:

BMS – STING Agonist: Characterized immune microenvironment using scRNA-seq + scTCR-seq in patient samples stratified by clinical treatment response (responder/non-responder); analyzed cell type composition dynamics across response cohorts to identify combination therapy-responsive clones and inform therapeutic prioritization. Published in Cell Reports Medicine (2025).

Takeda – TAK-676 + PD-1 Blocker: Characterized immune cell composition across six treatment cohorts; built automated reporting pipeline delivering wiki-style interactive outputs directly accessible to pharmaceutical partners. Presented at AACR 2024.

Takeda – NK Cell Therapy: Developed ML classifier (Random Forest, Gradient Boosting) to distinguish exogenous therapeutic from endogenous NK cells in unlabeled patient samples — leveraging phenotypic divergence as a biological prior. Predicted NK fractions validated against expected sample composition. Presented at AACR 2024.

Daiichi-Sankyo – DS-1062a + Pembrolizumab (NSCLC): Built end-to-end analysis pipeline integrating scRNA-seq with pseudo-bulk RNA-seq for immune biomarker identification across treatment conditions.

MGH Jain Lab – Anti-CD47 (PTCL): Delivered multi-omics framework (scRNA-seq, RNA-seq, ATAC-seq) linking chromatin accessibility to transcriptional programs for therapeutic target discovery. Contributed to experimental design as an embedded computational consultant across multiple projects.

Built modular pipeline infrastructure deployed across all four partnerships — reducing setup time and ensuring methodological consistency. Mentored junior scientists on single-cell analysis best practices

POSTDOCTORAL FELLOW | HACKENSACK MERIDIAN HEALTH, NUTLEY, NJ | 2021 – 2022

Center for Discovery & Innovation, Hai-Hui Xue Laboratory

Resolved transcriptional and epigenomic regulation of CD8+ T cell fate determination by integrating scRNA-seq/CITE-seq, ATAC-seq, and CUT&RUN. Developed domain knowledge-driven feature selection strategy using DEGs as biological anchors across chromatin datasets — computational outputs directly supported in vivo validation experiments. Published in Nature Immunology (2024, co-first author).

TECHNICAL SKILLS

Data Infrastructure & Pipeline Development: Multi-omics integration architecture (RNA-seq, ATAC-seq, CUT&RUN, proteomics, metabolomics, Cell Painting — up to 6 modalities); systematic benchmarking of integration methods (DIABLO, MOFA2, sparseCCA, custom design). Workflow engineering: Snakemake (production pipelines), conda, Git, HPC (PBS/SLURM), DNAnexus. Programmatic public data integration (TCGA, GTEx, GEO, CCLE); data standardization and QC framework design.

Programming & Software Development: Python (pandas, scikit-learn, NumPy), R (statistical computing, ggplot2, Shiny, R Markdown), Linux/bash scripting. Deep learning: PyTorch, Keras, TensorFlow (VAE/CVAE). AI-assisted development (GitHub Copilot, VS Code).

Single-Cell Genomics: scRNA-seq, scTCR-seq, CITE-seq — applied to target discovery, immune cell state characterization, and therapy response analysis. Expert in Seurat, Scanpy, Harmony, scVI, CellRanger, Monocle, DoubletFinder

Statistical Methods & Machine Learning Interpretable supervised learning: Random Forest, Gradient Boosting, Logistic Regression with feature importance-driven biological interpretation. Dimensionality reduction & clustering: PCA, t-SNE, UMAP; hierarchical, k-means, DBSCAN. Differential expression: DESeq2, edgeR, limma; pseudo-bulk approaches, cross-validation, ROC-AUC

Biological Domain Expertise: Deep expertise: immuno-oncology, neurodegeneration, metabolic disease (hepatic transcriptomics, high fat diet models). Rapid domain adaptation across physiology, cancer immunology, and defense applications.

Reproducibility & Documentation: Snakemake workflows, R Markdown, Jupyter Notebook; comprehensive methodology reporting and code annotation

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SELECTED PUBLICATIONS

1. The transcriptional cofactor Tle3 reciprocally controls effector and central memory CD8+ T cell fates. *Nature Immunology*, 2024. Co-first author.
2. Immune targeting of triple-negative breast cancer through a clinically actionable STING agonist-CAR T cell platform. *Cell Reports Medicine*, 2025.
3. Autism-associated ASPM variant causes macrocephaly and social-cognitive deficits in mice. *bioRxiv* [Preprint], 2025. *(Nature Neuroscience revision)*
4. Seq-Scope: Microscopic examination of spatial transcriptome. *Cell*, 2021
5. Single cell transcriptome analysis of colon cancer cell response to 5-fluorouracil-induced DNA damage. *Cell Reports*, 2020. First author.
6. Holistic Characterization of Single Hepatocyte Transcriptome Responses to High Fat Diet. *AJP-Endocrinology and Metabolism*, 2020. First author